data synchronisations across the Web application layer.

Hence, it is preferable to design a push-based and event-driven architecture pattern, where a chain of events propagates without any user input. These types of event-driven patterns give the serverless architecture its main power, which is scalability.

Identify performance bottlenecks

The key benefit of serverless architecture is that developers can concentrate on writing code rather than focusing on how and where to run the code. However, when it comes to functions, understanding the performance bottlenecks is a tricky subject. The absence of a familiar monitoring environment, compared to what developers were used to in the traditional system, makes it difficult for them to understand why a particular function is taking so long to execute.

For example, let's look at the case of functions calling other functions. Incorporation of such anti-patterns not only doubles overall cost but also makes the debugging more complex, while abstracting the isolation value from your functions. Ultimately, identifying the performance bottlenecks may take a while, but is highly critical for a seamless customer experience.

Incorporate standard security mechanisms

Functions are governed by different security policies. This makes it important to incorporate proper security mechanisms for both the API gateway and the functions. The IAM policy will include aspects like access controls, identity and access management, authentication management, encryption and decryption, and much more.

For example, you may violate this best practice by employing external libraries. Using external libraries will also result in more cold starts, since more libraries need more instantiating. In terms of security, more libraries equal more vulnerabilities. This makes it highly critical for you to follow the Principle of Least Privilege.

Use third party services

While you're developing a software application, try not to build every functionality on your own. Doing so might add a lot more complexities. In the serverless ecosystem, lately, exciting enterprise tools for various services like logging, monitoring, etc, have been developed.

The current tools that you are using might not be compatible considering the event-driven approach of serverless architecture. Choosing the right third party tools will be the key for enterprises to reap the maximum benefits from serverless architecture.

With the right processes in the architectural practice, serverless architecture offers your organisation an exit strategy from the legacy infrastructure, and enables you to make the most out of the public serverless offerings. Such architecture is still undergoing many changes, and will need to evolve a lot more before mainstream adoption happens on a large scale.

Till we have a standard set of processes to follow, incorporating the best practices can help you enjoy the basic advantages of serverless computing, i.e., reduced time-to-market, enhanced scalability and cost benefits. END

By: Kunjal Panchal

The author is a brand strategist at Simform, a custom software development company. She is passionate about content marketing and strongly believes in the power of storytelling for marketing. She has written for publications like Search Engine Journal, Entrepreneur, and many more. A perfect day for her involves reading her favourite author with a hot cuppa coffee.